Cardiovascular System (CARD) · NPLEX-P1-CARD-18

CV-18-J Vascular Neoplasms

Brain dumps can be right in more than one way. Grade the ideas and the relationships, not the wording. A response is competent when it carries the essentials below.

NPLEX-P1-CARD-18-10-C01-BD1

Brain dump 1 · foundational retrieval: Hemangioma against malformation, and the life of an infantile hemangioma

A competent response contains

· A hemangioma is a benign neoplasm of endothelial cells that forms blood-filled vascular channels; a vascular malformation is an error in vessel development, present at birth, that grows in proportion to the child and never regresses.

· Benign vascular tumors form well-formed channels lined by a single layer of bland endothelium; malignant ones form irregular channels lined by crowded atypical cells.

· Life cycle, in order: absent or faint precursor patch at birth; appears in the first weeks of life; GLUT1-positive endothelium proliferates rapidly for months; endothelial cells die by apoptosis and fibrofatty tissue replaces them; most have regressed by early childhood.

· Incidence is about 4 to 5% of infants, and it is the most common tumor of infancy.

· It is more common in girls and in preterm and low birth weight infants.

· Superficial lesions are bright red and raised; deep lesions are bluish.

· Lesional endothelium expresses GLUT1 in every phase, a marker otherwise typical of placental capillaries; vascular malformations and congenital hemangiomas, which are fully formed at birth, are GLUT1 negative.

· A capillary hemangioma is a well-circumscribed but unencapsulated lobule of closely packed thin-walled capillaries in skin, subcutaneous tissue or oral mucosa.

· A cavernous hemangioma is made of large dilated thin-walled channels filled with blood, separated by scant connective tissue, less well circumscribed, and often extending into deeper tissue.

· The infantile lesion regresses because its proliferating endothelium runs a programme that ends in apoptosis; the cavernous lesion does not regress because its vessels are mature rather than proliferating.

· Involution can leave lax skin or telangiectasias behind.

Distinctions that matter

· Proliferation against malformation is the dividing line: a hemangioma has dividing endothelial cells, a malformation has structurally abnormal vessels whose cells are not proliferating abnormally.

· Capillary and cavernous describe the calibre of the channels, not the timing of the lesion, so neither word tells you whether it will regress.

Common misconceptions

· Treating any red birthmark as a hemangioma. A lesion fully formed at birth that grows with the child is a malformation.

· Expecting cavernous lesions to involute like infantile ones. Their channels are mature, so they persist, and slow flow leads instead to thrombosis and calcification.

Where it is taught: Section C01, opening paragraphs, the life cycle flow and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-BD2

Brain dump 2 · integration and application: Site, growth rate and the harm they cause

A competent response contains

· Harm depends on where the lesion grows and how fast, not on malignancy.

· Periorbital lesion blocking the visual axis or compressing the globe: amblyopia.

· Beard distribution lesion: involvement of the subglottic airway, so airway obstruction.

· Large segmental facial lesion: part of PHACE syndrome, which is posterior fossa malformations, hemangioma, arterial anomalies, cardiac defects such as coarctation, and eye anomalies.

· Large hepatic lesion: it expresses type 3 iodothyronine deiodinase, which inactivates thyroid hormone and can cause consumptive hypothyroidism.

· Ulceration is the most common complication, because rapid growth outstrips the blood supply of the overlying skin.

· The newborn with a patch present at birth that grows in proportion to the child has a port-wine stain, a capillary malformation: GLUT1 negative, no regression, often in a trigeminal distribution, linked to Sturge-Weber syndrome with a somatic GNAQ mutation.

· Kasabach-Merritt phenomenon, the platelet-trapping consumptive coagulopathy, arises in kaposiform hemangioendothelioma and tufted angioma, both GLUT1 negative, and not in the ordinary infantile hemangioma.

Distinctions that matter

· Site decides the complication while growth rate decides ulceration, so two lesions of the same size can carry quite different risks.

· Kasabach-Merritt belongs to two specific GLUT1-negative lesions, which is why the GLUT1 result is worth knowing in a bleeding infant.

Common misconceptions

· Attaching Kasabach-Merritt phenomenon to the ordinary strawberry hemangioma.

· Expecting a port-wine stain to fade in childhood. It persists and in fact darkens and thickens with age.

Where it is taught: Section C01, the paragraph on complications, the cavernous paragraph and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-RT01

Retrieval exercise: Rebuild the life cycle of an infantile hemangioma

A competent response contains

· Step 1: absent or only a faint precursor patch at birth.

· Step 2: appears in the first weeks of life.

· Step 3: GLUT1-positive endothelium proliferates rapidly for months, then growth plateaus.

· Step 4: endothelial cells die by apoptosis and fibrofatty tissue replaces them.

· Step 5: most have regressed by early childhood.

· The marker is GLUT1, otherwise typical of placental capillaries.

· Regression is by apoptosis, with fibrofatty replacement, sometimes leaving lax skin or telangiectasias.

· The feature that marks this as a tumor rather than a malformation is proliferation after birth followed by programmed regression; a malformation is present at birth and never regresses.

Distinctions that matter

· Growth plateau and involution are two separate stages, and the plateau comes first.

· GLUT1 positivity is present in every phase, so a negative result argues against the infantile lesion even during proliferation.

Common misconceptions

· Writing that the lesion is present and fully formed at birth. At most there is a faint precursor patch.

· Describing involution as the vessels simply shrinking. The endothelial cells die and fibrofatty tissue takes their place.

Where it is taught: Section C01, the life cycle flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-CP01

Comparison table: Infantile hemangioma, cavernous hemangioma and port-wine stain

A competent response contains

· At birth: infantile lesion absent or a faint precursor; cavernous variable and often found later; port-wine stain present.

· Growth: infantile rapid for months then plateaus; cavernous slow or none; port-wine stain in proportion to the child, darkening and thickening with age.

· Regression: infantile yes over years; cavernous no; port-wine stain no.

· GLUT1: infantile positive; cavernous negative; port-wine stain negative.

· Histology: infantile lobules of small packed capillaries; cavernous large dilated blood-filled spaces; port-wine stain ectatic dermal capillaries and venules.

· Sites: infantile head and neck skin; cavernous skin, liver as the most common benign liver tumor, and brain; port-wine stain face, often in a trigeminal distribution.

· Associations: infantile lesion with PHACE when large and segmental on the face; cavernous with thrombosis and dystrophic calcification; port-wine stain with Sturge-Weber syndrome from a somatic GNAQ mutation.

Distinctions that matter

· Present at birth and never regressing means malformation; appearing after birth, growing and then fading means infantile hemangioma.

· A hepatic cavernous lesion is usually a small solitary incidental finding in an adult, which is a different clinical situation from a large hepatic infantile hemangioma in a baby.

Common misconceptions

· Filling the GLUT1 row as positive for all three. Only the infantile lesion is positive.

· Putting liver under the infantile row as the classic site. The liver is the classic site for the cavernous lesion as the most common benign liver tumor.

Where it is taught: Section C01, the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-BD1

Brain dump 1 · foundational retrieval: Three red papules that are not hemangiomas of infancy

A competent response contains

· Pyogenic granuloma: any age, often after minor trauma, and on the gingiva in pregnancy; a rapidly growing exophytic red polyp enlarging over weeks on the fingers, face, gingiva or oral mucosa; histology shows lobules of capillaries divided by fibrous septa in edematous stroma with inflammatory cells, resembling exuberant granulation tissue, with a collar of epidermis at the base and a thin frequently ulcerated surface; it bleeds with slight contact.

· Its histologic name is lobular capillary hemangioma, and the name pyogenic granuloma is inaccurate twice over because the lesion is neither infectious nor granulomatous.

· The pregnancy form is granuloma gravidarum, sitting on the gingiva; estrogen and progesterone amplify the vascular response of the gingiva to local irritants such as dental plaque, and it often regresses after delivery.

· Cherry hemangioma (senile angioma): small bright red dome-shaped papules on the trunk and proximal limbs, appearing from early adulthood and multiplying with age until nearly universal in older adults; histology shows dilated capillaries and venules in the papillary dermis; they do not regress and have no malignant potential.

· Bacillary angiomatosis: an infection rather than a neoplasm, in immunocompromised hosts and especially people with AIDS and very low CD4 counts; red to purple papules and nodules of skin, sometimes with bone, lymph node or liver involvement, including peliosis hepatis with B. henselae.

· The organisms are Bartonella henselae, from cats and their fleas, and Bartonella quintana, from the human body louse; both are small gram-negative bacilli.

· They induce host cells to produce hypoxia-inducible factor 1 alpha, which raises VEGF and drives capillary proliferation.

· Histology shows capillary proliferation with plump epithelioid endothelial cells, a neutrophilic infiltrate with nuclear dust, and clumps of granular purple material that a Warthin-Starry silver stain resolves into bacilli.

· Because living bacteria sustain the proliferation, the lesions disappear when the organism is eradicated.

Distinctions that matter

· Pyogenic granuloma is a reactive capillary proliferation, while cherry hemangioma is an age-related collection of dilated vessels and bacillary angiomatosis is an infection; only the infantile hemangioma in the same table is a true neoplasm driven from within the endothelium.

· Neutrophils with nuclear dust and silver-stain-positive bacilli mark bacillary angiomatosis, whereas the inflammatory cells of pyogenic granuloma sit in edematous stroma with no organisms.

Common misconceptions

· Reading the word pyogenic as evidence of infection and treating the lesion as an abscess.

· Expecting cherry hemangiomas to carry malignant potential or to regress. They persist and are benign.

Where it is taught: Section C02, the three descriptive paragraphs and the comparison table of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-BD2

Brain dump 2 · integration and application: A purple papule in a man with a CD4 count of 40

A competent response contains

· Both bacillary angiomatosis and Kaposi sarcoma fit violaceous skin papules in a man with a CD4 count of 40 cells per microliter, so the color of the lesions does not decide between them.

· Bacillary angiomatosis on biopsy: rounded capillaries, plump epithelioid endothelial cells, a neutrophilic infiltrate with nuclear dust, and silver-stain-positive bacilli on Warthin-Starry staining.

· Kaposi sarcoma on biopsy: fascicles of spindle cells around slit-like spaces with extravasated red cells, lymphocytes and plasma cells, and nuclear staining for LANA.

· Fever and systemic illness favour an infection, and the cat at home fits Bartonella henselae acquired from cats and their fleas.

· Bacillary angiomatosis resolves when the organism is eradicated, because living bacteria sustain the proliferation through hypoxia-inducible factor 1 alpha and VEGF.

· Kaposi sarcoma depends on immune recovery, because the latent viral genes keep driving the spindle cells until cytotoxic T cells and NK cells can again control the infected cells and the inflammatory cytokines that feed them.

Distinctions that matter

· Rounded capillaries with plump endothelium against spindle cells around slit-like spaces is the architectural difference, and it is more reliable than any clinical feature.

· One lesion needs the bacterium present and the other needs the immune system absent, which is why the two respond to quite different changes in the host.

Common misconceptions

· Deciding between the two from the color or distribution of the papules.

· Assuming both lesions will persist until immunity recovers. Bacillary angiomatosis disappears once the bacteria are cleared.

Where it is taught: Section C02, the bacillary angiomatosis paragraph and the worked example on a purple papule in a man with AIDS of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-CP02

Comparison table: Red and purple vascular lesions side by side

A competent response contains

· Infantile hemangioma: infant in the first weeks of life; an endothelial neoplasm; packed capillaries that are GLUT1 positive; grows then regresses.

· Pyogenic granuloma: any age, after trauma, or on the gingiva in pregnancy; a reactive capillary proliferation; lobular capillaries in edematous inflamed stroma with an ulcerated surface; bleeds easily, and the pregnancy form often regresses after delivery.

· Cherry hemangioma: adults, increasing with age; cause unknown and age related; dilated capillaries and venules in the papillary dermis; persists and is benign.

· Bacillary angiomatosis: immunocompromised hosts, especially advanced AIDS; Bartonella henselae or quintana; epithelioid endothelium with neutrophils and silver-stain-positive bacilli; resolves when the infection is eradicated.

· Kaposi sarcoma: older Mediterranean men, people with AIDS, transplant recipients, and the African endemic groups; HHV-8 with immune dysfunction; spindle cells with slit-like spaces, lymphocytes, plasma cells and hemosiderin; a low-grade malignancy whose course depends on immune status.

Distinctions that matter

· Neutrophils and bacteria point to bacillary angiomatosis; spindle cells and slit-like spaces point to Kaposi sarcoma.

· Only two of the five rows hold a true neoplasm of endothelium, and the other three are a reactive proliferation, an age-related ectasia and an infection.

Common misconceptions

· Grouping all five as tumors. The cause column is the point of the table.

· Expecting the course column to follow the appearance of the lesion rather than the cause.

Where it is taught: Section C02, the table of red and purple vascular lesions of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-BD1

Brain dump 1 · foundational retrieval: How HHV-8 drives the spindle cell

A competent response contains

· HHV-8 (Kaposi sarcoma herpesvirus) is a double-stranded DNA gammaherpesvirus, spread mainly through saliva and also by sexual contact, transplanted organs and blood.

· Infection is necessary but not sufficient: most infected people never develop the tumor, and the usual cofactor is impaired T-cell immunity.

· The tumor cell is the spindle cell, of endothelial lineage and largely lymphatic endothelial, expressing CD31 and CD34 along with podoplanin and PROX1.

· In most spindle cells the virus is latent, kept as a circular episome expressing only a few genes, each of which pushes the cell towards division or survival.

· Viral cyclin is a homolog of cyclin D: it binds CDK6, resists the CDK inhibitors p16, p21 and p27, and phosphorylates Rb, freeing E2F to drive the G1 to S transition.

· LANA tethers the viral episome to host chromosomes so daughter cells inherit it, and it binds and inhibits both p53 and Rb, removing the checkpoint that would arrest or kill a cell proliferating without a signal.

· Viral FLIP activates NF-kappaB and blocks apoptosis.

· A small fraction of cells enters lytic replication, secreting viral IL-6 and expressing a constitutively active viral G protein-coupled receptor that increases VEGF; these paracrine signals promote angiogenesis and inflammation and so recruit and stimulate neighbouring cells.

· Early lesions are polyclonal and advanced nodules may become clonal, which is why Kaposi sarcoma is described as lying between reactive hyperplasia and true neoplasia.

Distinctions that matter

· Latent genes supply proliferation and survival within the infected cell, while lytic genes supply paracrine signals that act on the neighbours.

· Viral cyclin phosphorylates Rb and LANA inhibits Rb and p53, so Rb is hit twice by two different mechanisms.

Common misconceptions

· Treating HHV-8 infection as sufficient to cause the tumor. Impaired T-cell immunity is the usual cofactor.

· Calling the spindle cell a fibroblast or smooth muscle cell. It carries endothelial and lymphatic endothelial markers.

Where it is taught: Section C03, the pathogenesis paragraphs and the flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-BD2

Brain dump 2 · integration and application: From slit-like spaces to the findings at the bedside

A competent response contains

· Patch: a flat pink, red or purple macule; histology shows irregular dilated vascular spaces with lymphocytes, plasma cells and macrophages, resembling granulation tissue.

· Plaque: a raised violaceous plaque; histology shows dermal spindle cells around jagged channels with extravasated red cells and hemosiderin-laden macrophages.

· Nodule: a purple to brown nodule that may ulcerate; histology shows sheets of plump spindle cells with slit-like spaces holding red cells and eosinophilic hyaline globules.

· Purple color: the slit-like channels are leaky, so red cells escape into the tissue.

· Brown color of old lesions: hemoglobin from those extravasated red cells is broken down to hemosiderin.

· Swelling of the limb: involvement of lymphatics and lymph nodes causes lymphedema, which fits the lymphatic endothelial lineage of the spindle cell.

· Bleeding from a gut lesion: mucosal and visceral lesions are made of leaky vascular spaces under a thin surface.

· Nuclear staining for LANA identifies HHV-8 in the spindle cells at every stage.

· When immunosuppression is lowered the lesions often shrink, because restored cytotoxic T-cell and NK cell activity kills infected spindle cells and reduces the inflammatory cytokines feeding them, even though the latent viral genes themselves have not changed.

· HHV-8 also causes primary effusion lymphoma and multicentric Castleman disease.

Distinctions that matter

· Color tracks the age of the lesion because it tracks the fate of the extravasated red cells, from purple to brown.

· Regression on restoring immunity shows that growth depends on the host as well as on the virus, which is the argument for calling the early lesion closer to hyperplasia than to neoplasia.

Common misconceptions

· Reading regression after reduced immunosuppression as loss of the virus. The virus persists; the control over it returns.

· Expecting LANA staining only in advanced nodules. It marks the spindle cells at every stage.

Where it is taught: Section C03, the staging table, the clinical findings paragraph, the immune control paragraph and the stop and predict box of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-RT01

Retrieval exercise: Rebuild the path from infection to a nodule

A competent response contains

· Step 1: HHV-8 infects endothelial-lineage cells and becomes latent.

· Step 2: viral cyclin phosphorylates Rb and LANA inhibits p53 and Rb.

· Step 3: viral FLIP activates NF-kappaB and blocks apoptosis.

· Step 4: lytic cells release viral IL-6 and raise VEGF.

· Step 5: weak T-cell and NK control lets infected cells expand.

· Step 6: spindle cells form leaky slit-like spaces.

· Step 7: patch progresses to plaque, then nodule.

· Steps 2 and 3 are the latent programme, step 4 is the lytic programme, and step 5 is the host immune system.

· Host proteins put out of action by the latent genes: Rb, p53, and the apoptotic pathway downstream of NF-kappaB activation; the CDK inhibitors p16, p21 and p27 are resisted.

· The lytic step releases viral IL-6 and the viral G protein-coupled receptor signal, and VEGF rises.

· The immune step depends on loss of control by cytotoxic T cells and NK cells.

· Latent genes give proliferation and survival, lytic genes give paracrine growth signals, and immune suppression removes the brake, so losing any one of the three stalls the sequence.

Distinctions that matter

· The latent and lytic programmes run in different cells at the same time, so both contribute to one lesion.

· Immune suppression is permissive rather than causative: it removes a brake instead of supplying a growth signal.

Common misconceptions

· Putting the immune step first as though immunosuppression caused the infection. Infection comes first and latency is established before the immune state matters.

· Assigning viral IL-6 and VEGF to the latent programme. They come from the small lytic fraction.

Where it is taught: Section C03, the flow diagram from infection to nodule of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-PR01

Prediction exercise: Predict what happens when single viral genes are lost

A competent response contains

· Viral cyclin inactivated: Rb phosphorylation falls, E2F stays bound, and the G1 to S transition is no longer forced.

· Viral cyclin inactivated: p16, p21 and p27 regain their braking effect, because it was the viral cyclin and CDK6 complex that resisted them.

· LANA lost: the episome is no longer tethered to host chromosomes, so daughter cells fail to inherit it and the infection is lost from the dividing population.

· LANA lost: p53 and Rb activity is restored, so a cell dividing without a signal is arrested or killed at the checkpoint.

· Viral FLIP lost: NF-kappaB signalling falls and apoptosis is no longer blocked, so infected cells die more readily.

· Lytic genes silenced: viral IL-6 and VEGF fall, angiogenesis and inflammation decline, and neighbouring cells lose the paracrine stimulus that recruited and stimulated them.

· Immune function restored with all viral genes intact: the lesions often shrink, because cytotoxic T cells and NK cells again kill latently infected cells and the inflammatory milieu that fed them subsides.

Distinctions that matter

· LANA does two separable jobs, one mechanical in keeping the episome and one regulatory in inhibiting p53 and Rb, so losing it costs the virus both persistence and checkpoint evasion.

· Removing a viral gene acts inside the infected cell, while restoring immunity acts on the infected cell from outside; both reduce the lesion but by different routes.

Common misconceptions

· Expecting restored p53 and Rb to have no effect because the cell is already transformed. Early lesions are polyclonal and still depend on these drivers.

· Assuming the lytic genes matter only for making new virus. Their products are the paracrine growth signals for the lesion.

Where it is taught: Section C03, the latent and lytic gene paragraphs, the immune control paragraph and the flow diagram of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C04-BD1

Brain dump 1 · foundational retrieval: Build the four forms of Kaposi sarcoma from memory

A competent response contains

· Every form contains HHV-8 and shows the same histology, so what differs is the host.

· Classic: older men, usually over 60, of Mediterranean, Eastern European or Ashkenazi Jewish descent and HIV negative; skin of the distal lower limbs with multiple plaques and nodules; indolent over years to decades with uncommon visceral spread.

· Endemic African: HIV-negative children and adults in sub-Saharan Africa, where most people acquire HHV-8 in childhood through saliva; adults have skin of the limbs and may have bone infiltration, children have lymph node disease; adults are locally aggressive and the childhood lymphadenopathic form is rapidly progressive.

· Transplant-associated (iatrogenic): solid organ recipients on T-cell immunosuppression, months to years after transplant; skin, mucosa, lymph nodes and viscera; can be aggressive and often regresses when immunosuppression is reduced.

· AIDS-associated (epidemic): people with HIV infection, mainly men who have sex with men; skin anywhere including face and trunk, oral mucosa and hard palate, gastrointestinal tract, lungs and lymph nodes; the most aggressive and widespread form, an AIDS-defining illness that improves with immune reconstitution.

· The two host variables are how common HHV-8 infection is in the population and how deeply T-cell immunity is suppressed.

· Mild immune decline gives slow disease confined to the skin of the legs; profound suppression gives multifocal disease of skin, mucosa, lymph nodes and viscera.

Distinctions that matter

· Population prevalence of the virus explains where the classic and endemic forms occur, while depth of immunosuppression explains how fast and how widely any form spreads.

· Transplant disease arises either from reactivation of the recipient's latent virus or from virus carried in the donor organ, which is a different route from the sexual and salivary spread behind the AIDS form.

Common misconceptions

· Expecting different histology for each form. The histology is the same and only the host differs.

· Treating the endemic African form as mild because the patients are HIV negative. The childhood lymphadenopathic form progresses quickly.

Where it is taught: Section C04, the opening paragraph, the table of forms and the four descriptive paragraphs of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C04-BD2

Brain dump 2 · integration and application: Four patients, four courses

A competent response contains

· The 74-year-old Mediterranean man has classic Kaposi sarcoma; the cofactor is age-related decline in T-cell function in a population where HHV-8 infection is relatively common, and the course over a year is slow growth of plaques and nodules on the feet and lower legs.

· The 6-year-old in sub-Saharan Africa has the endemic lymphadenopathic form; the cofactor is childhood acquisition of HHV-8 through saliva in a high-prevalence population, and the course is rapid progression even without skin lesions.

· The woman 14 months after kidney transplantation has transplant-associated disease; the cofactor is pharmacological T-cell suppression, and the course can be aggressive across skin, mucosa, lymph nodes and viscera.

· The man with HIV infection has AIDS-associated disease; the cofactor is CD4 depletion, and the course is the most aggressive and widespread, with pulmonary lesions causing dyspnea, hemoptysis and pleural effusions.

· The classic patient is the one most likely to die of an unrelated cause, because the disease is indolent over years to decades and visceral spread is uncommon.

· The man with HIV infection has the AIDS-defining illness.

· The transplant recipient's disease is the one most likely to shrink without treatment aimed at the tumor, because lowering immunosuppression restores T-cell control.

· The AIDS form is more common in men who have sex with men because HHV-8 spreads sexually and through saliva within those networks, which reflects transmission of the virus rather than a property of HIV itself.

· During early immune reconstitution lesions may flare transiently, because recovering immune cells increase local inflammation before overall control improves.

Distinctions that matter

· Pace and extent follow immune status, so the same virus and the same histology can give decades of indolent leg disease or widespread visceral disease within months.

· A flare during early immune recovery is inflammatory, not a sign that the tumor has escaped control.

Common misconceptions

· Reading the distribution of the AIDS form as evidence that HIV itself makes the tumor more aggressive in one group. The difference reflects how HHV-8 spreads.

· Expecting all four patients to need the same intensity of concern. The classic form rarely determines the outcome of the patient's life.

Where it is taught: Section C04, the table of forms and the paragraphs on each form of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-BD1

Brain dump 1 · foundational retrieval: Myxoma and rhabdomyoma from the ground up

A competent response contains

· Primary cardiac tumors are rare and metastases to the heart are far more common.

· Myxoma is the most common primary cardiac tumor in adults, usually appearing in middle age and more often in women.

· About 90% of myxomas arise in the atria, with the left atrium affected about four times as often as the right, typically from the interatrial septum near the fossa ovalis.

· Gross appearance ranges from a firm sessile mass to a soft gelatinous villous lesion on a stalk.

· Histology shows stellate or globular myxoma cells, thought to derive from primitive multipotent mesenchymal cells, in abundant acid mucopolysaccharide ground substance, often with hemorrhage and chronic inflammation.

· About 10% of myxomas are familial as part of Carney complex, an autosomal dominant disorder caused by loss-of-function mutations in PRKAR1A, a regulatory subunit of protein kinase A; these tumors are often multiple, recur, arise in younger patients and unusual sites, and come with spotty skin pigmentation and endocrine overactivity.

· Rhabdomyoma is the most common primary cardiac tumor of infants and children, a hamartoma of cardiac muscle cells, often multiple, within the ventricular myocardium.

· Rhabdomyoma is strongly associated with tuberous sclerosis, in which mutation of TSC1 (hamartin) or TSC2 (tuberin) removes inhibition of mTOR and allows excess cell growth; inheritance is autosomal dominant.

· Its large glycogen-filled myocytes have cytoplasm stretched in strands from a central nucleus to the cell membrane, the so-called spider cells, and many lesions regress spontaneously during early childhood.

Distinctions that matter

· Myxoma is a true neoplasm of primitive mesenchymal cells, while rhabdomyoma is a hamartoma of cardiac muscle, which is why only the rhabdomyoma regresses.

· Adult plus left atrium suggests myxoma; child plus multiple ventricular masses suggests rhabdomyoma and tuberous sclerosis.

Common misconceptions

· Calling the myxoma matrix fibrous. It is abundant acid mucopolysaccharide ground substance, which is why the tumor is gelatinous and friable.

· Treating the spider cell as a tumor cell of malignant type. It is a glycogen-filled cardiac myocyte within a hamartoma.

Where it is taught: Section C05, the myxoma and rhabdomyoma paragraphs and the table of primary cardiac tumors of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-BD2

Brain dump 2 · integration and application: Why a benign tumor makes a patient faint

A competent response contains

· Sequence: a stalked mass attached to the atrial septum; it prolapses into the mitral orifice in diastole; ball-valve obstruction to left ventricular filling; left atrial and pulmonary venous pressure rise; dyspnea, pulmonary edema and positional syncope; friable fragments break off as systemic emboli.

· The three groups of findings are obstruction, embolisation and constitutional symptoms.

· Obstruction mimics mitral stenosis with a diastolic murmur, and as the tumor drops into the ventricle there is a low-pitched early diastolic tumor plop; repeated trauma can damage the leaflets and cause mitral regurgitation.

· Obstruction varies with tumor position, so the murmur and the symptoms change with posture and the symptoms are episodic.

· A left atrial tumor sheds friable tumor or overlying thrombus into the systemic circulation, causing stroke or limb ischemia; a right atrial myxoma embolises to the lungs.

· Many myxomas secrete IL-6, which drives the hepatic acute phase response, raising ESR and CRP and producing anemia of inflammation, with fever, weight loss, fatigue and arthralgia.

· Echocardiography shows a mobile mass attached to the septum.

· A rhabdomyoma causes trouble by obstructing a valve orifice or outflow tract and by causing arrhythmias.

· Its problems differ because it sits within the ventricular myocardium rather than hanging free in an atrium, so it blocks flow from inside the wall and disturbs conduction instead of acting as a mobile ball valve or shedding emboli.

Distinctions that matter

· Obstruction from a mobile mass varies with position, while fixed valvular stenosis does not, which is what makes the symptoms postural.

· The constitutional symptoms come from a secreted cytokine, not from spread of a benign tumor.

Common misconceptions

· Explaining fever and a raised ESR in this setting as infection by default. IL-6 from the tumor produces the same acute phase picture.

· Assuming any atrial myxoma causes stroke. A right atrial tumor embolises to the lungs.

Where it is taught: Section C05, the flow diagram and the paragraph on the three groups of findings, with the rhabdomyoma paragraph of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C05-CA01

Clinical application: Breathlessness, fainting on standing and a high ESR

A competent response contains

· Recognize: a left atrial myxoma, attached to the interatrial septum near the fossa ovalis, and the sound after the second heart sound is a low-pitched early diastolic tumor plop.

· Explain: in diastole the stalked mass prolapses into the mitral orifice and obstructs left ventricular filling as a ball valve, so left atrial and pulmonary venous pressures rise and give exertional breathlessness; because obstruction depends on where the tumor sits, a change of position can block filling abruptly, cardiac output falls, and she faints.

· Differentiate: rheumatic mitral stenosis gives fixed obstruction that does not change with posture and would not explain the fever, weight loss, high ESR and anemia, and there is no history of rheumatic fever; infective endocarditis is argued against by repeatedly negative blood cultures, and it would not produce a mobile mass on the atrial septum with a tumor plop.

· Investigate: echocardiography, which shows a mobile mass attached to the interatrial septum and its movement into the mitral orifice in diastole; the raised ESR and the raised CRP are the laboratory findings already present that support the systemic picture, alongside the anemia.

· Decide: embolisation of friable tumor fragments or overlying thrombus is the most damaging complication, and from a left atrial tumor the brain is the organ most likely to be affected, giving stroke.

· Justify: many myxomas secrete IL-6, which drives the hepatic acute phase response, so fever, weight loss, a high ESR and anemia of inflammation all follow from one secreted cytokine; negative blood cultures do not argue against the diagnosis because the systemic picture is cytokine driven rather than infective.

Distinctions that matter

· Postural change in the murmur and in the symptoms separates a mobile mass from fixed valvular stenosis.

· A benign tumor is dangerous here through mechanics and embolisation rather than through invasion or metastasis.

Common misconceptions

· Reading fever, weight loss and a high ESR with a new murmur as endocarditis without weighing the mobility of the lesion and the negative cultures.

· Expecting a benign cardiac tumor to be harmless. Obstruction, emboli and IL-6 all cause harm without any malignant behavior.

Where it is taught: Section C05, the flow diagram, the paragraph on the three groups of findings and the worked example on posture-dependent syncope of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-BD1

Brain dump 1 · foundational retrieval: Angiosarcoma: what it is, where it sits, what it looks like

A competent response contains

· Angiosarcoma is a malignant neoplasm of endothelial cells, occurring mainly in older adults.

· It can arise anywhere, most often in the skin and especially the scalp and face of elderly patients, and in soft tissue, breast and liver; it is the most common primary malignant tumor of the heart in adults, usually in the right atrium.

· Well-differentiated tumors form irregular anastomosing channels that dissect between collagen bundles, lined by plump atypical multilayered endothelial cells.

· Poorly differentiated tumors grow as solid sheets of anaplastic spindle or epithelioid cells with mitoses and necrosis, so endothelial markers are needed to show the lineage; the two markers named are CD31 and ERG.

· Risk factors: vinyl chloride in PVC production workers, main site liver; arsenic from pesticides, smelting or contaminated well water, main site liver with skin and lung cancers as well; Thorotrast, a radiocontrast used about 1930 to 1950, main site liver with cholangiocarcinoma and hepatocellular carcinoma; chronic lymphedema in the arm after radical mastectomy with axillary dissection, which is Stewart-Treves syndrome, arising in the lymphedematous limb; prior breast or chest wall irradiation, arising in the irradiated field; and no identified exposure in older adults, arising in the scalp and face skin.

· Behavior: it infiltrates widely beyond its visible margin, is often multifocal in the skin, recurs locally and metastasizes frequently to the lungs.

· Five-year survival is about 30%, and hepatic tumors do worse.

Distinctions that matter

· Channel formation separates well from poorly differentiated tumors, and once channels are lost only the endothelial markers establish the lineage.

· The hepatic risk factors act through the liver itself, while lymphedema and radiation act on a damaged field.

Common misconceptions

· Expecting a clear surgical margin from the visible edge. The tumor infiltrates well beyond what can be seen.

· Reading a solid sheet of anaplastic cells as excluding a vascular tumor. CD31 and ERG settle the lineage.

Where it is taught: Section C06, the opening paragraph, the risk factor table and the final paragraph on behavior of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-BD2

Brain dump 2 · integration and application: Same tumor, two different organs

A competent response contains

· Both patients have angiosarcoma, a malignant endothelial tumor, but the injury differs, so the organ differs.

· Sequence in the PVC worker: inhaled vinyl chloride monomer in PVC manufacturing; hepatic CYP2E1 oxidizes it to chloroethylene oxide; the epoxide forms DNA adducts in sinusoidal cells; mutations accumulate over decades of latency; malignant endothelial proliferation arises in the liver.

· The liver rather than the lung is the target because the metabolic activation step happens there, so the reactive epoxide is generated next to hepatic sinusoidal cells.

· The woman has Stewart-Treves syndrome, angiosarcoma arising about a decade after chronic lymphedema of the arm following radical mastectomy with axillary node dissection.

· The two features of a chronically lymphedematous limb thought to allow the tumor are impaired local immune surveillance and growth factor rich lymphedematous tissue.

· Angiosarcoma arising in a previously irradiated breast or chest wall field often carries MYC amplification.

· The shared property is a long latent period, often decades, between the exposure and the tumor.

· The other two hepatic exposures are arsenic, from pesticides, industrial exposure or contaminated well water, and Thorotrast, a thorium dioxide contrast agent used from about 1930 to 1950 that was taken up by hepatic and splenic macrophages and retained for life, irradiating nearby cells with alpha particles.

Distinctions that matter

· A chemical that needs metabolic activation picks the organ that activates it, whereas a field injury picks the field, which is why the same tumor appears in liver or limb.

· Thorotrast and radiation both injure by ionizing radiation, but one is retained inside hepatic macrophages for life and the other is delivered to an external field.

Common misconceptions

· Expecting vinyl chloride to cause lung tumors because it is inhaled. The target follows the site of metabolic activation.

· Treating the long gap between exposure and tumor as evidence against a causal link. Decades of latency are typical for all of these exposures.

Where it is taught: Section C06, the two risk factor paragraphs, the vinyl chloride flow diagram, the risk factor table and the stop and predict box of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C01-Q01

Board-style question: A red lesion that appeared after birth

Correct answer: A. Rapid growth for several months, then apoptosis of the endothelium with fibrofatty replacement

Why it is right: Infantile hemangioma proliferates rapidly for several months, then growth plateaus and the endothelial cells die by apoptosis and are replaced by fibrofatty tissue, so most lesions have regressed by early childhood; preterm birth, female sex, the faint precursor patch, the lobules of packed capillaries and GLUT1 staining all fit this lesion.

B is wrong: Growth in proportion to the child with darkening and thickening in adult life describes a port-wine stain, which is present at birth and GLUT1 negative.

C is wrong: Progressive dilatation of large thin-walled channels with thrombosis and calcification describes a cavernous hemangioma, whose mature vessels do not proliferate and which is GLUT1 negative.

D is wrong: Infiltration of deep tissue with metastasis to the lungs describes angiosarcoma, a malignant tumor of older adults with atypical multilayered endothelium rather than bland packed capillaries.

A competent response contains

· A faint precursor patch at birth, appearance in the first weeks of life, steady enlargement, lobules of packed capillaries and GLUT1-positive endothelium together identify an infantile hemangioma.

· The natural history is rapid proliferation for months, a plateau, then apoptosis of the endothelium with fibrofatty replacement over years.

Distinctions that matter

· GLUT1 positivity separates the infantile hemangioma from vascular malformations and congenital hemangiomas, which are GLUT1 negative.

Common misconceptions

· Reading rapid postnatal growth as a sign of malignancy. Proliferation followed by programmed regression is the expected behavior here.

Where it is taught: Section C01 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C02-Q02

Board-style question: Purple nodules in an immunocompromised host

Correct answer: C. Bacterial induction of hypoxia-inducible factor 1 alpha with a rise in VEGF

Why it is right: Bartonella henselae, acquired here from kittens, induces host production of hypoxia-inducible factor 1 alpha, which raises VEGF and drives capillary proliferation, and the neutrophilic infiltrate with Warthin-Starry positive bacilli is the histological signature of that infection.

A is wrong: A viral cyclin homolog binding CDK6 and phosphorylating Rb in spindle cells is the mechanism of Kaposi sarcoma, which would show spindle cells around slit-like spaces and no bacilli.

B is wrong: A somatic activating GNAQ mutation in dermal endothelium belongs to the port-wine stain and Sturge-Weber syndrome, a capillary malformation present from birth.

D is wrong: Endothelial type 3 iodothyronine deiodinase expression belongs to large hepatic infantile hemangiomas and causes consumptive hypothyroidism rather than a neutrophilic vascular proliferation.

A competent response contains

· Rounded capillaries with plump epithelioid endothelium, neutrophils with nuclear dust and silver-stain-positive bacilli in an immunocompromised host identify bacillary angiomatosis.

· Bartonella induces host cells to produce hypoxia-inducible factor 1 alpha, which raises VEGF and drives the capillary proliferation, so the lesions resolve when the organism is eradicated.

Distinctions that matter

· Neutrophils and visible bacilli point to bacillary angiomatosis, while spindle cells around slit-like spaces point to Kaposi sarcoma in the same patient population.

Common misconceptions

· Assuming any purple nodule in advanced HIV infection is Kaposi sarcoma. The inflammatory cells and the vessel shape decide it.

Where it is taught: Section C02 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C03-Q03

Board-style question: The gene that removes two checkpoints at once

Correct answer: B. It binds and inhibits both p53 and Rb, removing the checkpoint response to unscheduled proliferation

Why it is right: LANA is the latency-associated nuclear antigen detected by the nuclear stain; besides tethering the viral episome to host chromosomes so daughter cells inherit it, it binds and inhibits both p53 and Rb, so the checkpoint that would arrest or kill an inappropriately dividing cell is lost.

A is wrong: Activating NF-kappaB to block apoptosis is the action of viral FLIP, not of the tethering nuclear antigen.

C is wrong: A constitutively active G protein-coupled receptor that raises VEGF is expressed by the small lytic fraction of cells and acts as a paracrine signal rather than on the episome.

D is wrong: Resisting the inhibitors p16, p21 and p27 is a property of the viral cyclin and CDK6 complex.

A competent response contains

· The nuclear antigen that tethers the HHV-8 episome to host chromosomes is LANA, latency-associated nuclear antigen.

· LANA also binds and inhibits both p53 and Rb, removing the checkpoint that would arrest or kill a cell proliferating without a signal.

Distinctions that matter

· Each latent gene has its own job: viral cyclin drives the cell cycle, LANA keeps the episome and disables p53 and Rb, and viral FLIP blocks apoptosis.

Common misconceptions

· Attributing all of the latent effects to one protein. The question turns on which protein does which job.

Where it is taught: Section C03 of the CV-18-J Vascular Neoplasms notes.

NPLEX-P1-CARD-18-10-C06-Q04

Board-style question: A liver mass decades after an industrial exposure

Correct answer: D. Hepatic cytochrome P450 2E1 oxidized the monomer to a reactive epoxide that formed DNA adducts in sinusoidal cells

Why it is right: Vinyl chloride monomer is oxidized by hepatic cytochrome P450 2E1 to chloroethylene oxide, a reactive epoxide that forms DNA adducts in sinusoidal cells, so mutations accumulate in hepatic endothelium over decades and the malignant proliferation arises in the liver.

A is wrong: Retention of particles by hepatic macrophages with alpha particle irradiation describes Thorotrast, a thorium dioxide contrast agent, not an inhaled monomer from PVC manufacturing.

B is wrong: Impaired lymphatic drainage in a growth factor rich tissue describes the mechanism proposed for Stewart-Treves angiosarcoma in a chronically lymphedematous limb.

C is wrong: MYC amplification is characteristic of secondary angiosarcoma arising in a previously irradiated breast or chest wall field.

A competent response contains

· Irregular anastomosing channels dissecting between collagen bundles with plump multilayered atypical CD31-positive cells in the liver of a former PVC worker identify hepatic angiosarcoma from vinyl chloride.

· Hepatic CYP2E1 oxidizes vinyl chloride to chloroethylene oxide, a reactive epoxide that forms DNA adducts in sinusoidal cells, which is why the liver is the target organ.

Distinctions that matter

· Metabolic activation in the liver explains the organ for vinyl chloride, while lymphedema and radiation explain the field for the other classic exposures.

Common misconceptions

· Expecting an inhaled carcinogen to strike the lung. The site of activation, not the route of entry, picks the organ here.

Where it is taught: Section C06 of the CV-18-J Vascular Neoplasms notes.